

The Cathedral and Society: Legal, Policy, and Educational Perspectives on Machine-Verified Mathematics

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Abstract

We examine the legal, policy, and educational implications of the Cathedral project. We address intellectual property questions (can a machine-verified proof be patented? who owns AI-assisted mathematics?), governance implications (how should formally verified results be treated by standards bodies?), and educational applications (how can the Cathedral be used to teach analytic number theory, formal verification, and mathematical research methodology?).

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1 Introduction

The Cathedral is a mathematical object, but it exists in a legal, political, and educational context. A formally verified reduction of the Riemann Hypothesis raises questions that pure mathematics does not usually encounter.

2 Intellectual Property

2.1 Copyright

Mathematical theorems cannot be copyrighted (Baker v. Selden, 1879). However, the *expression* of a proof—the specific Lean code, the paper text, the documentation—is copyrightable. The Cathedral’s code is released under an open-source license.

2.2 Patents

Mathematical methods are generally not patentable (Alice Corp. v. CLS Bank, 2014). However, specific *applications* of the Cathedral’s mathematical infrastructure—such as the Möbius Harmonic Analyzer or the arithmetic fingerprinting protocol—could potentially be patented as methods or devices.

The author has chosen not to pursue patents, consistent with the tradition of mathematical openness.

2.3 AI Authorship

The Cathedral was built with significant AI contributions. Under current U.S. law (Thaler v. Perlmutter, 2023), AI systems cannot be listed as authors or inventors. The human author retains sole authorship, with AI contributions acknowledged.

This legal framework may evolve. The Cathedral’s detailed records of human-AI collaboration (COMM-LINKs, exploration logs) provide a useful case study for future legal analysis.

3 Policy Implications

3.1 Standards for Formal Verification

Should formally verified mathematical results carry more weight than traditional peer-reviewed proofs? We argue yes, with caveats:

1. **Compiler trust:** The verification is only as trustworthy as the Lean kernel (5,000 LOC).
2. **Specification trust:** The formalization must faithfully represent the mathematical intent.
3. **Axiom transparency:** All axioms must be explicitly documented.

The Cathedral satisfies all three criteria.

3.2 Dual-Use Governance

The Cathedral’s dual-use paper (*The Dark Mirror*) identifies 7 risk categories where the mathematical infrastructure could be weaponized. This raises governance questions:

- Should verified instability certificates be classified?
- Who is responsible for misuse of openly published mathematical results?
- How should the tension between mathematical openness and security be managed?

We follow the tradition of responsible disclosure: publishing the mathematics with explicit threat analysis and defense strategies.

3.3 Funding and Incentives

Formal verification is expensive (45 days of intensive work for the Cathedral). Current academic incentive structures reward traditional papers, not machine-verified proofs. Policy changes that could help:

- Funding agencies accepting formal proofs as deliverables.
- Journals accepting Lean code as supplementary material.
- Tenure committees recognizing formalization contributions.

4 Educational Applications

4.1 Teaching Analytic Number Theory

The Cathedral provides a structured curriculum:

1. **Week 1–2:** The Nyman–Beurling criterion (definitions, Gram matrix, positive definiteness).
2. **Week 3–4:** The converse direction (Mellin transforms, rank-1 factorization).
3. **Week 5–6:** The forward direction (Parseval bridge, Hardy–Littlewood MVT).
4. **Week 7–8:** Numerical experiments (Rust/MPFR, Vasyunin formula, eigenvalue computation).

Each topic has both a mathematical treatment (the companion papers) and a machine-verified formalization (the Lean code).

4.2 Teaching Formal Verification

The Cathedral demonstrates practical ITP skills:

- Working with Mathlib APIs (interval integrals, matrices).
- Managing axioms in evolving codebases.
- The layered module architecture pattern.
- Debugging type-class inference failures.

4.3 Teaching Research Methodology

The Cathedral’s archive of 35 explorations, including abandoned approaches, 5 formalization traps, and 20+ axiom graduations provides a realistic picture of mathematical research—including the failures that are usually invisible in published papers.

5 Accessibility

The Cathedral is designed for accessibility:

- All code is open source.
- All papers are freely available.
- The Lean code compiles with standard tooling.
- The Rust experiments build with `cargo build`.
- No proprietary software is required at any step.

6 Conclusion

The Cathedral exists at the intersection of mathematics, computer science, law, and education. Its open-source release, transparent axiom documentation, and dual-use risk assessment provide a model for responsible publication of formally verified frontier mathematics.

References

- [1] J.R. Gochanour, *The Cathedral*, 2026.
- [2] Alice Corp. v. CLS Bank International, 573 U.S. 208 (2014).
- [3] Thaler v. Perlmutter, No. 1:22-cv-01564 (D.D.C. 2023).